DG INTERN HUB

SOC ANALYST INTERNSHIP

WEEK 2 —NOTES

Submitted By: Gunjan

Domain: Cybersecurity / SOC Analyst

Platform: TryHackMe

Submission Date: 15 October 2026

Alert, Log Entry and Malware Behaviour Mapping to MITRE ATT&CK

Learning Objective

To understand how a SOC analyst can investigate suspicious activity and map it to the correct MITRE ATT&CK tactic and technique.

---

1. How does a SOC analyst map an alert to MITRE ATT&CK?

When a SOC analyst receives an alert, they first need to understand what actually happened instead of directly deciding that it is malicious.

The analyst can follow these basic steps:

1. **Check the alert details**
    - What happened?
    - Which system generated the alert?
    - When did it happen?
    - Which user was involved?
2. **Look at the related logs**
    - Process logs
    - Windows Event Logs
    - Network logs
    - Authentication logs
    - EDR/SIEM alerts
3. **Understand the behaviour**
    - Was a command executed?
    - Was a new account created?
    - Was a suspicious file downloaded?

    - Was there network scanning?
    - Was a user sent a phishing email?
4. **Search MITRE ATT&CK**
    - Find the technique that matches the observed behaviour.
    - Check the technique description and examples.
5. **Identify the tactic**
    - After identifying the technique, check which tactic it belongs to.
6. **Use the mapping during investigation**
    - Look for other events that could be related to the same attack.
    - Check whether the attacker may have used additional techniques.

---

2. Example: Suspicious PowerShell Activity

Suppose a SOC analyst receives an alert from an endpoint security tool.

The alert shows:

powershell.exe -ExecutionPolicy Bypass -File update.ps1

The analyst should not immediately say that this is definitely an attack because PowerShell can also be used for normal administrative work.

The analyst can investigate:

- Which user executed PowerShell?
- What was the parent process?
- What is inside `update.ps1`?
- Was the script downloaded from the internet?
- Did PowerShell make any network connections?
- Were there other suspicious events around the same time?

If the behaviour is confirmed to be PowerShell execution, it can be mapped to:

Item	*Mapping*

Observed behaviour	*PowerShell command execution*
Tactic	*Execution*
Technique	*Command and Scripting Interpreter: PowerShell*
Evidence	*Process/command-line logs*
Investigation	*Check user, command, parent process, file and network activity*

This mapping gives the SOC analyst more information about what the attacker may be trying to do.

---

3. Another Example: Phishing Email

Suppose an employee receives an email that looks like it is from a bank.

The email contains a link to a fake login page.

The analyst can investigate:

- Sender address
- Sender domain
- Email headers
- URL
- Domain reputation
- Whether the user clicked the link
- Whether the user entered credentials

If the email is confirmed to be malicious, the behaviour can be mapped to:

Tactic: Initial Access
Technique: Phishing

The analyst can then check if the same sender, domain or URL was seen in other emails.

---

4. Why is ATT&CK mapping useful?

Mapping activity to MITRE ATT&CK can help a SOC analyst:

- Understand attacker behaviour
- Connect different alerts
- Identify possible attack stages
- Improve threat hunting
- Create better detection rules
- Communicate findings in a common format
- Identify gaps in security monitoring

For example, if several alerts are related to different techniques, the analyst may be able to understand that they are part of the same attack instead of treating each alert separately.

---

5. Basic Investigation Flow

I can think of the process like this:

Alert / Log

↓

Understand the activity

↓

Collect related evidence

↓

Identify attacker behaviour

↓

Search MITRE ATT&CK

↓

Identify Technique

↓

Identify Tactic

↓

Continue investigation

---

6. My Understanding

I understood that mapping an alert to MITRE ATT&CK is not just about searching for a keyword and selecting a technique.

First, the analyst needs to understand what actually happened and collect enough evidence.

For example, if I see PowerShell in a log, I should not immediately mark it as malicious. I need to check what command was executed, who executed it, what started the process and whether there was any other suspicious activity.

After understanding the behaviour, I can map it to the relevant MITRE ATT&CK technique and tactic.

This makes the investigation more structured and helps me understand the bigger picture of an attack.